Increased Inequality in Financial Wealth for Retirees*

Kaja Kierulf†

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Abstract

Inequality in financial wealth for retirees increased in the U.S. between 1989 and 2019; the P90/P10 ratio rose from 7.48 to 11.13, an increase of 3.64. This paper uses a discrete-time, partial equilibrium, overlapping generations life cycle model with heterogeneous agents, calibrated to the U.S. economy, to study how three channels contributed to this increase. Each channel's contribution is measured as the difference in inequality between letting it vary over time and holding it constant at its 1973/74 level. The shift from defined benefit toward defined contribution plans and the introduction of IRAs raises the P90/P10 ratio by 0.19, rising earnings inequality by 0.84, and rising, increasingly heterogeneous life expectancy by 0.18. The full model increases the ratio by 1.58, about 43 percent of the increase in the data. It reproduces the direction but not the full magnitude.

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†E-mail: kieru001@umn.edu.

1 Introduction

Inequality in financial wealth for retirees increased in the U.S. between 1989 and 2019. Financial wealth is measured as the sum of taxable assets net of debt, defined contribution and individual retirement account assets, and present value of expected Social Security and defined benefit payments. The $P90/P10$ ratio, the 90th percentile divided by the 10th percentile, increased from 7.48 to 11.13. This yields an increase in the ratio of 3.64 over the time period. This paper asks how changes in retirement plans, earnings, and life expectancy have contributed to this increase.

Unlike workers, retirees have no labor income; their financial wealth determines both their consumption in retirement and the resources they transmit to the next generation. Understanding what drives the inequality in financial wealth for this group therefore matters for retirement security, intergenerational mobility, and the design of retirement and Social Security policy. Central to the paper is the shift from defined benefit to defined contribution plans, which transferred the decision of how much to save for retirement from employers to workers themselves.

I use a life cycle model with heterogeneous agents to study the question. In addition to age, individuals are heterogeneous in six states: education, idiosyncratic earnings shock, average life cycle earnings, type of retirement plan, tax-deferred assets, and standard assets. *Ex ante*, individuals have either college or non-college education. An individual's survival probability depends on cohort, age, and education. A worker's earnings depend on an idiosyncratic earnings shock in addition to their age, level of education, and time. Social Security benefits depend on the individual's average life cycle earnings. An individual is either a defined benefit (DB), defined contribution (DC), or an individual retirement account (IRA) individual. An individual's possibility to take advantage of tax benefits when saving towards retirement depends on their type of retirement plan. All workers can save in a standard non-contingent taxable asset. Defined contribution and IRA workers can save a limited amount in a tax-deferred asset.

The workers' options to save towards retirement depend on their retirement plan. Defined benefit workers are subject to mandatory saving: A specified percent of their earnings is subtracted every period. In retirement they receive a predetermined benefit from the retirement plan every period they are alive. They cannot choose the amount saved in the retirement plan and they cannot save in the tax-deferred asset. The defined contribution workers can save a limited amount in the tax-deferred asset. They will receive a proportional match of their saving in this asset. In retirement they choose how much to dissave from the tax-deferred asset every period. An IRA worker is similar to a defined contribution worker. However, they do not receive any match to their tax-deferred saving. There are tax benefits to all the retirement plans.

I calibrate the model to the U.S. economy and use it to study the increase in inequality over the time period. The change in inequality is incorporated by exogenously changing over time the parameters governing each channel: change in retirement plans, increased inequality in earnings, and increased life expectancy and increased heterogeneity in life expectancy. In particular, the change in retirement plans is accounted for by changing the share of workers with different types of retirement plans. Increase in inequality in earnings is modeled by increasing the variance of the idiosyncratic earnings shock, increasing the college premium, and increasing the share of workers with college education. Increase in life expectancy and increase in heterogeneity in life expectancy are accounted for by changing the survival probability for college and non-college individuals. I measure the effect of each channel by comparing the change in inequality in financial wealth when either allowing the channel to vary over the time period as observed in the data or by keeping it at a constant level.

This paper yields three main findings. First, the change in the distribution of retirement plans and the introduction of IRA plans increase the $P90/P10$ ratio by 0.19. Financial wealth is more unequally distributed among defined contribution and IRA workers than among defined benefit workers, so the shift toward these plans raises inequality in financial wealth for retirees. Second, increased inequality in earnings increases the $P90/P10$ ratio by 0.84. Because a retiree's financial wealth reflects their earnings over the working life, wider earnings inequality produces wider inequality in financial wealth among retirees. Third, the increase in life expectancy and in its heterogeneity increases the $P90/P10$ ratio by 0.18. A worker who expects to live longer saves more for retirement, so rising and increasingly unequal life expectancy widens inequality in financial wealth among retirees.

The paper is related to three strands of literature. First, several papers have documented the increase in wealth inequality when including assets in retirement plans and Social Security. Devlin-Foltz, Henriques, and Sabelhaus (2016) uses SCF data from 1989 to 2013 and finds that wealth inequality for households increased once retirement plan assets are included. Ghilarducci, Radpour, and Webb (2022) uses HRS data from 1992 to 2010 and finds that inequality in retirement plan assets increased over the time period. Jacobs et al. (2021) uses SCF data from 1989 to 2019 and finds that wealth inequality, including assets in retirement plans and Social Security, increased for households aged 40 to 59. My contribution to this literature is to update the measurement of financial wealth and to restrict the sample to fully retired men. This sample is aligned with the life cycle model developed in the next section and allows me to isolate the mechanisms through which retirement plans, earnings, and life expectancy affect inequality.

Second, the paper is related to the literature on saving for retirement. Scholz, Seshadri, and Khitatrakun (2006) and De Nardi and Yang (2014) study retirement saving behavior in life cycle models, while İmrohoroğlu, İmrohoroğlu, and Joines (1998), Kitao (2010), Ho (2017), Love (2007), and Nishiyama (2011) develop life cycle models with tax-deferred retirement accounts. The closest paper in this strand is O'Dea (2018), who also models both defined contribution and defined benefit plans, but for the U.K. economy. Relative to this literature, I focus on the implications of the U.S. retirement plan system for inequality in financial wealth among retirees over time.

Third, the paper is related to the literature on wealth inequality. Huggett (1996), De Nardi (2004), Cagetti and De Nardi (2006), and De Nardi, French, and Jones (2010) study the sources of wealth inequality using heterogeneous agent life cycle and overlapping generations models. The closest paper in this strand is Brendler, Kuhn, and Steins (2024), who study how retirement plans contribute to wealth inequality through the extensive margin of retirement plan participation. Relative to this literature, I focus on both the extensive and intensive margin of saving in defined contribution and IRA accounts, include defined benefit plans in the model, and study the transition of inequality among retirees over time.

The rest of the paper is organized as follows. Section 2 documents the evidence of increased inequality in financial assets for retirees. Section 3 presents the benchmark model used for the analysis. The calibration is described in section 4. Section 5 presents the results and section 6 concludes.

2 Data

I document the increase in inequality in financial wealth for retirees by analyzing data from the Survey of Consumer Finances over the time period 1989-2019. The sample used in the analysis is men, at least 66 years old, who are fully retired.

Financial wealth is measured as financial assets net of debt. Financial assets include retirement assets and taxable financial assets. Retirement assets include assets in defined contribution plans and IRAs in addition to present value of expected Social Security and defined benefit payments. Expected Social Security and defined benefit payments are converted to present values by discounting future payments at the interest rate r, weighted by cohort-, age-, and education-specific survival probabilities. Taxable assets include bank accounts, bonds, stock, mutual funds, etc. The measure of debt includes lines of credit, credit cards, installment loans etc. Debt includes all measures of debt except debt related to real estate. While information on Social Security benefits, defined benefit payments, defined contribution plans and IRAs are given at the individual level, information on taxable assets and debt is given at the household level. If the household consists of a couple, I assume the man owns half of the net taxable assets.

Figure 1 displays the $P90/P10$ and $P75/P25$ ratios. The $P90/P10$ ratio is calculated by dividing the $90th$ percentile of the distribution by the $10th$ percentile of the distribution. It increased from 7.48 to 11.13, an increase of 3.64. The $P75/P25$ ratio is constructed by the corresponding calculations. Both the ratios increase over the time period which also indicates an increase in inequality.

Further, I decompose financial wealth into four parts: present value of expected Social Security benefit, present value of expected defined benefit payments, defined contribution and IRA assets, and taxable assets. Any debt is subtracted from the taxable assets in this calculation. The decomposition of the average financial wealth is displayed in figure 2a. The total value of average retirement wealth has increased over the time period. The graph

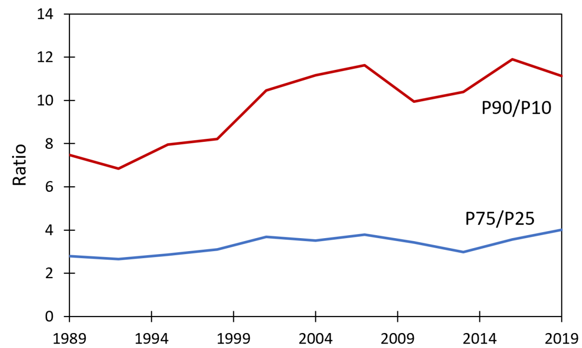


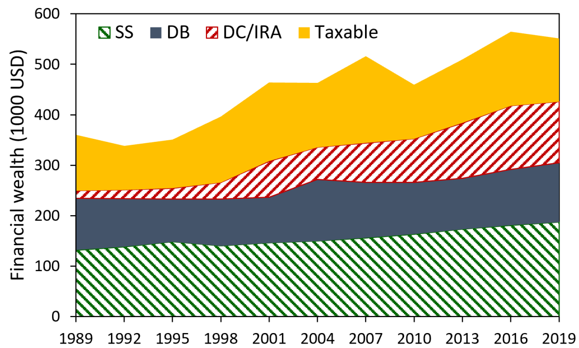
Figure 1: Ratios of 90/10 and 75/25 percentiles of financial wealth.

shows that Social Security, defined benefit and taxable assets are important components of financial wealth throughout the whole time period. However, defined contribution and IRA assets were not an important component of financial wealth in 1989, but the value of these assets increased substantially over the time period.

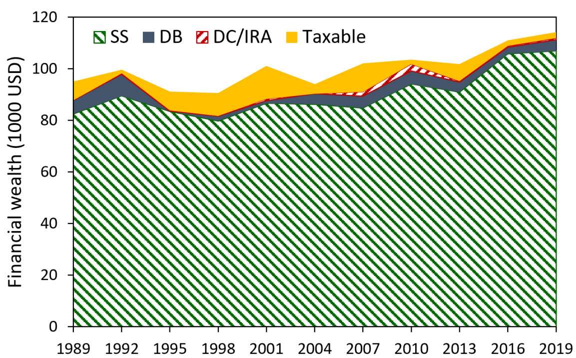
The decomposition of financial wealth is also computed for the 5-15 percentiles and 85-95 percentiles of the distribution. The decomposition of the 5-15 percentiles is displayed in figure 2b. Financial wealth increases slightly in the time period for this group. Almost all of their financial wealth consists of expected Social Security benefits. The decomposition of the 85-95 percentiles is displayed in figure 2c. Financial wealth increases for this group. The sources of wealth are more diverse for the 85-95 percentiles than for the 5-15 percentiles. Social Security assets increase slightly and are a small share of financial wealth. The value of defined benefit assets are roughly constant over the time period while taxable assets have increased slightly over the time period. Defined contribution and IRA assets have increased substantially over the time period.

This paper focuses on how three factors contribute to the increase in inequality in financial wealth for retirees: the introduction of IRAs and defined contribution retirement plans into the economy, the increase in earnings inequality, and the change in life expectancy. The rest of this section discusses how these factors changed over time.

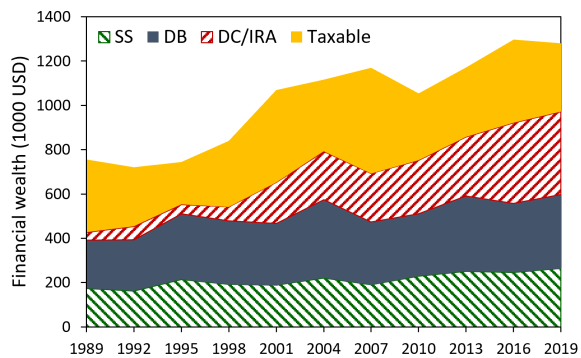
IRAs were authorized in the Employee Retirement Income Security Act (ERISA) of 1974. Workers could start saving in IRAs the year after, see Munnell (1985). The Revenue Act of 1978 added section 401(k) to the Internal Revenue Code. The act authorized the retirement



(a) Average



(b) Percentile 5-15



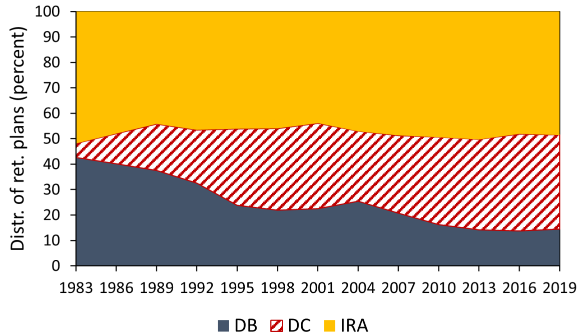
(c) Percentile 85-95

Figure 2: Decomposition of financial wealth for retirees into Social Security (SS), defined benefit (DB), defined contribution and IRA (DC/IRA), and taxable (Taxable) assets. The graphs show different parts of the wealth distribution. Wealth is measured in 2019 dollars.

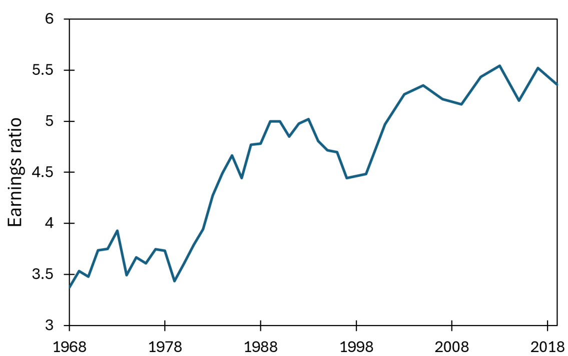
plans named 401(k) and they have become the most common type of defined contribution plan. The retirement plans were not widely adopted until after the IRS published regulations for the plans in 1981, see Congressional Research Service (2009). The type of retirement plans utilized in the economy changed after these policies were introduced. Figure 3a shows the distribution of the type of retirement plans from 1983 to 2019. The share of defined benefit plans decreased and the share of defined contribution plans increased over the time period. The share of workers with IRA retirement plans barely changed over the time period. The data is from the Survey of Consumer Finances and the sample used is men aged 25-65 who are working.

The increase in earnings inequality over the time period is documented in figure 3b. The figure shows the $P90/P10$ for men who are working. The data is from the PSID and the sample is men age 25-65.

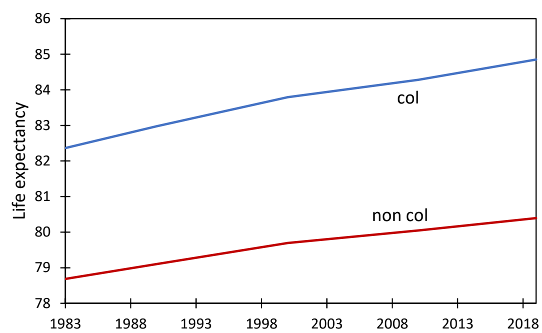
Life expectancy has increased for both college and non-college groups over the time period. The college population was expected to live longer than the non-college population at the beginning of the time period. Over the time period the college population increased the life expectancy more than the non-college population, i.e. the heterogeneity in life expectancy increased over the time period. These three factors motivate the model developed in the next section, which is designed to quantify their relative contributions to the increased inequality in financial wealth for retirees.



(a) The distribution of type of retirement plans for workers.



(b) The $P90/P10$ earnings ratio.



(c) Life expectancy for college and non-college respondents at age 65.

Figure 3: Factors contributing to increased inequality in financial wealth for retirees.

3 Model

The model used in the analysis is a discrete-time, partial equilibrium, overlapping generations model with heterogeneous agents.

3.1 Preliminaries

The economy is populated by a continuum of individuals. Each time period, t, a new cohort is born. The cohort size increases proportionally by g_n each period.

Individuals are indexed by type $s = \{j, e, \eta_t, y_t, \nu_t, d_t, a_t\}$, where j is age, e is education level, η_t is the stochastic component of labor productivity, y_t is average annual earnings, ν_t is the type of retirement plan, d_t is tax-deferred assets, and a_t is standard assets. Individuals of age j_R or higher are called retirees, and individuals younger than j_R are called workers. The maximum lifespan of an individual is J. The probability of survival until the next period, ψ_{jet}, depends on age, education, and the cohort the individual was born into. $\Phi_t(s)$ denotes the measure of individuals of type s at time t.

The individuals' education level is either college or non-college. It is set exogenously and is constant throughout their life. The share of college and non-college individuals born each period depends on time. There are three types of individuals in the economy: defined benefit (DB), defined contribution (DC), and individual retirement account (IRA) individuals. The share of each type of individual depends on age, education, and time.

Individuals derive utility from the consumption of the good, c_t, and are assumed to have standard time-separable preferences. The period-by-period utility function takes a constant relative risk aversion (CRRA) form:

$$U(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}.$$

Following De Nardi (2004), individuals derive utility from leaving bequests. The total bequest consists of the sum of standard and tax-deferred assets and yields utility:

$$\varphi(a_{t+1} + d_{t+1}) = \varphi_1 \left[(a_{t+1} + d_{t+1} + \varphi_2)^{1-\sigma} - 1 \right].$$

The parameter φ_1 governs the strength of the bequest motive, and φ_2 governs the extent to which bequests are a luxury good. A strictly negative φ_1 and a strictly positive φ_2 shift the function such that the utility of leaving bequests is bounded from below at zero wealth. For individuals with low wealth, the marginal utility of leaving bequests relative to consumption is greatly reduced. On the other hand, for individuals with high wealth, the marginal utility of leaving bequests is largely unaffected by the shifter.[1]

3.2 Earnings, Average Annual Earnings, Social Security Benefits, Taxes and Technological Growth

Individuals receive earnings from age 1 until age $j_R - 1$. Following Heathcote, Storesletten, and Violante (2010), earnings depend on a deterministic life cycle component, w_{jet}, and an individual-specific history of a stochastic component, η_t.

The deterministic life cycle component depends on age, education, and time. Due to aggregate technological growth, this component grows at rate γ each time period. For non-college workers, an individual with deterministic earnings component w_{jet} at age j at time t will receive the deterministic earnings component $w_{j+1et+1} = (1+\gamma)w_{j+1et}$ in the next time period. This captures both their change along the age profile and the aggregate technological growth. For college workers, the deterministic life cycle component additionally incorporates a time-varying education premium. The earnings component for a college worker of age j at time t is given by $\lambda_t w_{jet}$, where $\lambda_t > 0$ is a factor that is common across ages and varies over time.

The stochastic component of earnings follows an autoregressive process:

$$\eta_t = \rho\eta_{t-1} + \epsilon_t.$$

The white noise, ϵ_t, is drawn from a distribution with mean zero and time-varying variance $\sigma^2_{\epsilon t}$. The initial value of the stochastic shock at age 1 is drawn from a time-invariant distribution with mean zero and variance σ^2_{η}.

U.S. Social Security benefits are based on average earnings over the life cycle, subject to a maximum limit. Earnings before age 60 are adjusted for technological growth up to age 60, see Social Security Administration (2026). In the model, average annual earnings are

1. The utility function is non-homothetic due to the bequest motive and a balanced growth path does not exist. Details regarding the implementation of technological change follow in the next section.

indexed to technological growth. Specifically, earnings for each cohort are adjusted to the evaluation age, j_{SS}, which corresponds to age 60. Average annual earnings, y_t, is updated each period while working:

$$y_{t+1} = \frac{(j-1)y_t + \min\{\bar{y}_{t-j+j_{SS}}, (\eta_t w_{jet} - \xi_{st})(1+\gamma)^{j_{SS}-j}\}}{j}.$$

Only earnings up to the limit, $\bar{y}_{t-j+j_{SS}}$, are included in this measure. This upper limit is constant for each cohort, but grows at rate γ across cohorts such that $\bar{y}_{t+1} = (1+\gamma)\bar{y}_t$. The average annual earnings is calculated net of the retirement plan fee, ξ_{st}.

Beginning at age j_R, all individuals stop working and begin receiving Social Security benefits, SS_{jyt}, which depend on their average annual earnings and cohort. The benefits are modeled as in Huggett and Ventura (2000) and are a piecewise linear function of average annual earnings evaluated at the time of retirement, $t - j + j_R$:

$$SS_{jyt} = \begin{cases} \mu_1 y & \text{for } y \leq b^L_{t-j+j_R} \\ \mu_1 b^L_{t-j+j_R} + \mu_2(y - b^L_{t-j+j_R}) & \text{for } b^L_{t-j+j_R} < y \leq b^H_{t-j+j_R} \\ \mu_1 b^L_{t-j+j_R} + \mu_2(b^H_{t-j+j_R} - b^L_{t-j+j_R}) + \mu_3(y - b^H_{t-j+j_R}) & \text{for } b^H_{t-j+j_R} < y \leq \bar{y}_{t-j+j_R}. \end{cases} \tag{1}$$

The bendpoints, b^L_t and b^H_t, grow at the rate of technological change such that $b^L_{t+1} = (1+\gamma)b^L_t$ and $b^H_{t+1} = (1+\gamma)b^H_t$. The upper limit, $\bar{y}_t$, caps the average annual earnings and therefore the Social Security benefits. The marginal replacement rates are denoted by μ_1, μ_2, and μ_3. They decrease with average earnings ($\mu_1 > \mu_2 > \mu_3$) ensuring that Social Security benefits are progressive. Because the bendpoints and upper limit increase with aggregate technological growth, the progressivity of the benefit schedule is invariant to average annual earnings that grow at this same rate. By evaluating the bendpoints at time $t - j + j_R$, the benefits grow with technological change across cohorts but remain constant in retirement.

There are two types of taxes in the model: An income tax and the Federal Insurance Contribution Act (FICA) tax. Following Heathcote, Storesletten, and Violante (2017), the income tax is progressive and applies to both earnings and capital income with distinct parameters for workers and retirees. Let M_t be taxable income at time t. After-tax income for workers and retirees is $(1-\tau^W_0)M_t^{1-\tau^W_1}$ and $(1-\tau^R_0)M_t^{1-\tau^R_1}$, respectively. To adjust for technological growth, the tax function for workers at time $t+n$ is scaled to $(1+\gamma)^{n\tau_1^W}(1-\tau_0^W)M_t^{1-\tau_1^W}$. An analogous adjustment is applied to the retiree tax function. The FICA tax, τ^{FICA}, is a proportional tax subject to an earnings ceiling, $\bar{y}_t$, which grows at the rate of technological change.

3.3 Retirement Plans

An individual is classified as either a defined benefit, a defined contribution, or an IRA individual. An individual's type is set exogenously. Contributions to the retirement plans are tax-advantaged.[2] For defined benefit workers, contributions to the retirement plan are compulsory. For defined contribution and IRA workers, contributions act as saving in the tax-deferred asset and are optional. Defined benefit and IRA workers face a probability of becoming a defined contribution worker next period. Upon reaching retirement, an individual's plan type cannot change. Workers were not allowed to save in the tax-deferred asset before the policy change introducing defined contribution and IRA retirement plans into the economy.

A defined contribution worker chooses how much to save in the tax-deferred asset, x_t, each time period. The amount he chooses to save is matched proportionally by κ^{DC} and added to the tax-deferred asset. The defined contribution worker pays a lump-sum fee, ξ_{st}, every time period:

$$\xi_{st} = \kappa^{DC} x_t^* \qquad \text{if } \nu = DC,$$

where x_t^* is the worker's optimal saving choice. The fee is equal to the matching contribution such that the pre-tax value of the retirement plan each time period is zero. However, allowing for a proportional matching contribution and lump-sum fee increases the incentives for saving in the tax-deferred asset. The worker takes the fee as given when solving the optimization problem. A worker can save, but not dissave, from the tax-deferred asset. There are limits to how much the worker is allowed to save in the tax-deferred asset, and these limits are not

2. The tax treatment in the model follows IRS guidelines. See: https://www.irs.gov/retirement-plans/retirement-plan-faqs-regarding-contributions-are-retirement-plan-contributions-subject-to-withholding-for-fica-medicare-or-federal-income-tax.

equal to the corresponding limits for an IRA worker. A retiree cannot save, but can dissave from the tax-deferred asset. Each period, the retiree must dissave at least as much as the required minimum distributions, which depend on age. The tax-deferred assets accumulate tax-free. The workers do not pay income tax, but they do pay FICA tax on the amount, x_t, they save in the tax-deferred asset. The workers pay neither income tax nor FICA tax on the matching contribution, which is equal to the fee ξ_{st}. Retirees pay income tax, but not FICA tax, when they dissave in retirement. A defined contribution individual will remain a defined contribution individual for the rest of their life.

An IRA worker chooses how much to save in the tax-deferred asset, x_t, each time period. The worker does not receive a matching contribution on the investment and does not pay any fees:

$$\xi_{st} = 0 \qquad \text{if } \nu = IRA.$$

The incentives to save in the tax-deferred asset compared to the standard asset are due solely to differences in tax treatment. An IRA worker can save, but not dissave, from the tax-deferred asset. There are limits to how much the worker is allowed to save in the tax-deferred asset, and these limits are not equal to the corresponding limits for a defined contribution worker. A retiree cannot save, but can dissave from the tax-deferred asset. Every period, the retiree must dissave at least as much as the required minimum distributions, which depend on age. The tax-deferred assets accumulate tax-free. IRA workers do not pay income tax, but they do pay FICA tax on the amount, x_t, they save in the tax-deferred asset. Retirees pay income tax, but not FICA tax, when they dissave in retirement.

An IRA worker faces a probability, π_{jet}^{IRA}, of becoming a defined contribution worker next period. This probability depends on age, education, and time. If the worker transitions to a defined contribution worker, the amount of tax-deferred asset, d_t, does not change. The individual will remain a defined contribution individual for the rest of their life.

A defined benefit retiree receives a constant payment, $DB_{je\eta t}$, every time period. This payment depends on the last earnings they received, and therefore it depends on education, the labor productivity shock in the last period of work, and cohort. While working, a fee is subtracted from the worker's earnings every period. The fee, ξ_{st}, is a constant share of earnings:

$$\xi_{st} = \kappa_{jet}^{DB} \eta_t w_{jet} \qquad \text{if } \nu = DB.$$

The retirement plan is actuarially fair for each education and cohort group. This determines the share of earnings, κ_{jet}^{DB}, workers contribute to the retirement plan. For the cohort born at time $\underline{t}$, the share of earnings, κ_{jet}^{DB}, is:

$$\kappa_{jet}^{DB} = \frac{\sum_{\tilde{j}=j_R}^{J} \sum_{\eta} \frac{\prod_{\hat{j}=j_R-1}^{\tilde{j}} \psi_{\hat{j}e\hat{t}}}{(1+r)^{\tilde{j}-(j_R-1)}} DB_{je\eta t}}{\sum_{\tilde{j}=1}^{j_R-1} \sum_{\eta} \frac{(1+r)^{(j_R-1)-\tilde{j}}}{\prod_{\hat{j}=\tilde{j}}^{j_R-2} \psi_{\hat{j}e\hat{t}}} \eta_{\tilde{t}} w_{\tilde{j}e}}$$

$$\text{s.t.} \quad t = \underline{t} + j - 1, \quad \tilde{t} = \underline{t} + \tilde{j} - 1, \quad \hat{t} = \underline{t} + \hat{j} - 1.$$

The share of earnings, κ_{jet}^{DB}, depends on education and cohort only, and is therefore constant throughout the worker's life. The fee ensures that the ex-ante expected discounted value of the retirement plan pre-tax is equal to zero for each education and cohort group. The assets in the retirement plan accumulate tax-free. Defined benefit workers do not pay income tax or FICA tax on the fee used to fund the retirement plan. In retirement, they pay income tax on the payment they receive from the retirement plan, but they do not pay FICA tax. A defined benefit worker cannot save in the tax-deferred asset, d_t.

A defined benefit worker faces a probability, π_{jet}^{DB}, of becoming a defined contribution worker next period. This probability depends on age, education, and time. If the worker transitions to a defined contribution worker, they will start the period with a positive value of the tax-deferred asset, d_t. The value of the tax-deferred asset is an approximation of the accumulated value of the worker's contributions to the retirement plan:

$$d_t = \frac{y_t}{1 - \kappa_{jet}^{DB}} \kappa_{jet}^{DB} (1 + r) \frac{(1 + r)^{j-1} - 1}{r}.$$

Average annual earnings, y_t, is based on earnings after the contribution to the retirement plan. Dividing by $(1 - \kappa_{jet}^{DB})$ yields an approximation of earnings before the fee is subtracted. Altogether, this expression calculates the accumulated value of an annuity with a recurring payment equal to an approximated value of the worker's contributions to the retirement plan each period.[3] The individual will remain a defined contribution individual for the rest of their life and will therefore not receive any defined benefit payments in retirement.

3. Average annual earnings is an arithmetic average of earnings before FICA tax. The average includes

3.4 Worker and Retiree's Problems

Recall that individuals are indexed by type $s = \{j, e, \eta_t, y_t, \nu_t, d_t, a_t\}$, where j is age, e is education level, η_t is the stochastic component of earnings, y_t is average annual earnings, ν_t is type of retirement plan, d_t is tax-deferred assets, and a_t is standard assets. A worker

earnings up to the maximum value of earnings subject to the FICA tax. There are three reasons the value of tax-deferred assets is an approximation of the value of contributions to the defined benefit plan: First, earlier years of contributions to the defined benefit retirement plan accumulate returns for a longer time period than later years of contributions. Due to the earnings shock, some workers will have larger and some will have smaller contributions in the first years of their working life than their average annual earnings. The average annual earnings does not contain the exact history of the earnings shock for each individual, and it is therefore an approximation of the value of the accumulated contributions to the defined benefit retirement plan by each individual. Although this introduces idiosyncratic error at the individual level, it cancels out in the aggregate. Second, the deterministic component of earnings increases with age for young workers. On average, the fee contributed early is therefore smaller than the fee contributed later in the working life. The contributions earning the longest compounded interest are therefore on average smaller than later contributions. The approximated value for tax-deferred assets is therefore slightly larger than it would have been with the full history of contributions. Third, for high earners, average annual earnings is smaller than their actual earnings due to the maximum taxable value of the FICA tax. The approximated value for tax-deferred assets is therefore slightly lower than it would have been without capping the earnings for high earners.

solves the following problem:

$$
\begin{aligned}
V_t(j, e, \eta_t, y_t, \nu_t, d_t, a_t) = & \max_{c_t, x_t, a_{t+1}} U(c_t) \\
& + \beta \psi_{jet} E_{\eta_{t+1}|\eta_t t} E_{\nu_{t+1}|j\nu_t t} V_{t+1}(j+1, e, \eta_{t+1}, y_{t+1}, \nu_{t+1}, d_{t+1}, a_{t+1}) \\
& + (1 - \psi_{jet}) \varphi(d_{t+1} + a_{t+1}) \\
\text{s.t.} \qquad c_t = & (1 - \tau_0^W)\left(\eta_t w_{jet} - \xi_{st} - x_t + r a_t\right)^{1-\tau_1^W} \\
& - \tau^{FICA} \min\{\bar{y}_t, \eta_t w_{jet} - \xi_{st}\} + a_t - a_{t+1} \\
d_{t+1} = & x_t(1 + \mathbb{1}_{\nu_t = DC} \kappa^{DC}) + (1+r) d_t \\
y_{t+1} = & [(j-1) y_t + \min\{\bar{y}_{t-j+j_{SS}}, (\eta_t w_{jet} - \xi_{st})(1+\gamma)^{j_{SS}-j}\}]/j \\
x_t \in [0, \bar{x}_t], \quad \bar{x}_t = & \begin{cases} 0 & \text{for } \nu_t = DB \\ \min\{\hat{x}_t^{DC}, \theta^{DC} \eta_t w_{jet}\} & \text{for } \nu_t = DC \\ \min\{\hat{x}_t^{IRA}, \theta^{IRA} \eta_t w_{jet}\} & \text{for } \nu_t = IRA \end{cases} \\
a_{t+1} \geq 0, & \quad c_t \geq 0.
\end{aligned}
\tag{2}
$$

Each period, a worker chooses how much to consume, c_t, save in tax-deferred assets, x_t, and the level of standard assets next period, a_{t+1}, to maximize utility, V_t. Workers discount next period's utility by β. The probability of surviving until next period, ψ_{jet}, depends on age, education, and time. If the worker dies, they receive utility from bequeathing their assets, $\varphi(d_{t+1} + a_{t+1})$.

Workers receive earnings, $\eta_t w_{jet}$, each time period. Income tax is progressive and governed by the parameters τ_0^W and τ_1^W. The fee of the retirement plan, ξ_{st}, is deducted from the worker's income as a lump sum. Saving in tax-deferred assets is deducted from income before tax while saving in standard assets is deducted after tax. The Federal Insurance Contribution Act (FICA) tax, τ^{FICA}, is a flat tax up to a maximum of earnings, $\bar{y}_t$. This tax applies to earnings after the fee of the retirement plan is subtracted. The average annual earnings, y_t, is based on earnings after the fee of the retirement plan is subtracted, and only earnings up to the maximum limit, $\bar{y}_t$, are included in the average. Tax-deferred assets accumulate tax-free. The worker pays tax on returns on standard assets each period. The interest rate for both the tax-deferred and standard assets is r. Individuals cannot borrow in either the tax-deferred or standard asset.

When saving in the tax-deferred asset, defined contribution workers receive a matching contribution, κ^{DC}, on their savings. The indicator function $\mathbb{1}_{\nu_t=DC}$ equals one for defined contribution workers. Workers cannot dissave from the tax-deferred assets, and there is a maximum limit on how much they can save. θ^{DC} and θ^{IRA} are the maximum shares of earnings workers are allowed to save in the tax-deferred asset for defined contribution and IRA workers respectively. $\hat{x}_t^{DC}$ and $\hat{x}_t^{IRA}$ are the absolute maximum limits workers are allowed to save in the tax-deferred asset for defined contribution and IRA workers respectively. Taking both limits into account, $\bar{x}_t$ is the maximum limit the worker is allowed to save in tax-deferred assets. Workers were not allowed to save in the tax-deferred assets before the policy changes introducing IRA and defined contribution retirement plans. Defined benefit workers are not allowed to save in the tax-deferred asset.

A retiree solves the following problem:

$$\begin{aligned} V_t(j,e,\eta,y,\nu,d_t,a_t) &= \max_{c_t,x_t,a_{t+1}} U(c_t) + \beta\psi_{jet}V_{t+1}(j+1,e,\eta,y,\nu,d_{t+1},a_{t+1}) \\ &\quad + (1-\psi_{jet})\varphi(d_{t+1}+a_{t+1}) \\ \text{s.t.} \qquad c_t &= (1-\tau_0^R)\left(SS_{jyt} + \mathbb{1}_{\nu=DB}DB_{je\eta t} - x_t + ra_t\right)^{1-\tau_1^R} + a_t - a_{t+1} \\ d_{t+1} &= x_t + (1+r)d_t \\ x_t &\in [-(1+r)d_t, -\underline{\theta}_j(1+r)d_t] \\ a_{t+1} &\geq 0, \quad c_t \geq 0. \end{aligned} \tag{3}$$

Each period, the retiree chooses how much to consume, c_t, dissave from the tax-deferred assets, x_t, and the level of standard assets next period, a_{t+1}, to maximize utility, V_t. The retiree discounts next period's utility by β. The probability of surviving until next period, ψ_{jet}, depends on age, education, and time. If the retiree dies, they receive utility from bequeathing their assets, $\varphi(d_{t+1}+a_{t+1})$. The stochastic component of earnings, η, average annual earnings, y, and the type of retirement plan, ν, remain constant after the last period of work. These values are used to determine Social Security and defined benefit payments.

All retirees receive Social Security benefits, SS_{jyt}. The indicator function $\mathbb{1}_{\nu=DB}$ equals one for defined benefit retirees. These retirees receive payments, $DB_{je\eta t}$, which depend on education, the stochastic component of earnings, and cohort. Retirees pay tax on Social Security benefits, defined benefit payments, dissaving of tax-deferred assets, and return on standard assets. The tax is progressive and governed by τ_0^R and τ_1^R. The interest rate for both the retirement and standard assets is r. Individuals cannot borrow in either the tax-deferred or standard asset.

Defined contribution and IRA retirees cannot save in the tax-deferred asset. They must dissave at least as much as the required minimum distribution, $\underline{\theta}_j(1+r)d_t$. The share of the required minimum distribution, $\underline{\theta}_j$, increases with age. Defined benefit individuals cannot save in the tax-deferred assets while working, have zero tax-deferred assets, and therefore cannot dissave from the assets in retirement.

3.5 Equilibrium Definition

Let $s = (j, e, \eta_t, y_t, \nu_t, d_t, a_t)$ denote the individual state vector where j is age, e is education level, η_t is the stochastic component of earnings, y_t is the average annual earnings, ν_t is the type of retirement plan, d_t is the tax-deferred assets, and a_t is the standard assets. Given the retirement plan parameters $\{\kappa_{jet}^{DB}, \kappa^{DC}, \theta^{DC}, \theta^{IRA}, \hat{x}_t^{DC}, \hat{x}_t^{IRA}, \underline{\theta}_j\}$, the tax parameters $\{\tau_0^W, \tau_1^W, \tau_0^R, \tau_1^R, \tau^{FICA}\}$, the sequence of Social Security benefits $\{SS_{jyt}\}$, the exogenous interest rate r, the sequence of deterministic earnings components $\{w_{jet}\}$, a partial equilibrium is a sequence of value functions $\{V_t\}$, a sequence of decision rules $\{c_t, x_t, a_{t+1}\}$, and a sequence of probability measures describing the aggregate state of the economy $\{\Phi_t\}$, such that for all t:

1. Individual's problem: The value functions V_t and the associated policy functions $\{c_t, x_t, a_{t+1}\}$ solve the optimization problem (2) for an individual of age $j < j_R$ and the optimization problem (3) for an individual of age $j \geq j_R$.

2. Aggregate law of motion: For all individual states $s = (j, e, \eta_t, y_t, \nu_t, d_t, a_t) \in \mathcal{S}$ and any subset of the state space $S = (S_j \times S_e \times S_\eta \times S_y \times S_\nu \times S_d \times S_a) \in \Sigma_{\mathcal{S}}$, where $\{1\} \notin S_j$, Φ_t satisfy $\Phi_{t+1}(S) = \int_{\mathcal{S}} Q_t(s, S) d\Phi_t(s)$ where the transition function $Q_t\,(s, S)$ is given by:

$$Q_t\,(s, S) = \mathbb{1}_{\{j+1 \in S_j, e \in S_e, y_{t+1} \in S_y, d_{t+1} \in S_d, a_{t+1} \in S_a\}} \Pr_t\{\eta_{t+1} \in S_\eta \mid \eta_t\} \Pr_t\{\nu_{t+1} \in S_\nu \mid e, \nu_t\} \psi_{jet}.$$

The initial measure at age $j = 1$ for time t is

$$\Phi_t(\{1\}, S_e, S_\eta, \{0\}, S_\nu, \{0\}, \{0\}) = \\ (1+g_n)^t \Pr_t\{e \in S_e \mid j = 1\} \Pr\{\eta_t \in S_\eta \mid j = 1\} \Pr_t\{\nu_t \in S_\nu \mid j = 1, e\},$$

where g_n denotes the population growth rate.

4 Calibration

The model is matched to the data by determining some parameters outside the model equilibrium while others are calibrated to match moments in the U.S. economy. Estimates are based on men. The model period is 2 years. Parameters reported here are annualized, with a few clear exceptions. 1 unit in the model is 50 000 USD in 2019.

4.1 Parameters determined outside the model equilibrium

Table 1 reports parameters set outside the model equilibrium.

Demography, preferences and interest rate

Individuals enter the economy at age 25, retire at age 65 and can maximum live until age 100. Given a model period of 2 years, this implies the retirement age in the model j^R to be 21 and maximum lifespan J to be 38. The population growth g^n is 1.2 percent annually. Interest rate r is 4 percent annually. The coefficient of relative risk aversion is σ is 1.5.

Retirement plans

The matching contribution from firms for defined contribution workers ϕ_{DC} is set to 0.76. Maximum allowed absolute saving in the retirement asset for an IRA worker $\hat{x}_t^{\mu IRA}$ is 0 for years before 1975 and 0.091 thereafter. Maximum allowed absolute saving in the retirement asset for defined contribution worker $\hat{x}_t^{\mu DC}$ is 0 for years before 1982 and 0.343 thereafter. Maximum saving in the retirement asset relative to earnings for a defined contribution worker θ_{DC} and an IRA worker θ_{IRA} is set to 0.06 and 1.0 respectively.

Table 1: Parameters determined outside the model equilibrium

Parameter	Description	Value
j^R	Retirement age (65 years)	21
J	Maximum life span (100 years)	38
g^n	Population growth rate	0.012
r	Interest rate	0.04
σ	Risk aversion parameter	1.5
κ^{DC}	Matching rate from firms	0.76
$\hat{x}_t^{\mu DC}$	Max absolute saving ret. asset DC, 1982 and later	0.343
$\hat{x}_t^{\mu IRA}$	Max absolute saving ret. asset IRA, 1975 and later	0.091
θ^{DC}	Max relative saving ret. asset DC	0.06
θ^{IRA}	Max relative saving ret. asset IRA	1.0
ρ	Autocorrelation coefficient	0.971
σ_η^2	Variance of initial persistent shock	0.179
τ_0^W	Level tax parameter worker	0.066
τ_1^W	Progressive tax parameter worker	0.106
τ_0^R	Level tax parameter retiree	0.058
τ_1^R	Progressive tax parameter retiree	0.073
τ^{FICA}	FICA tax rate	0.153

Table 2: Distribution of retirement plans when cohorts enter the model.

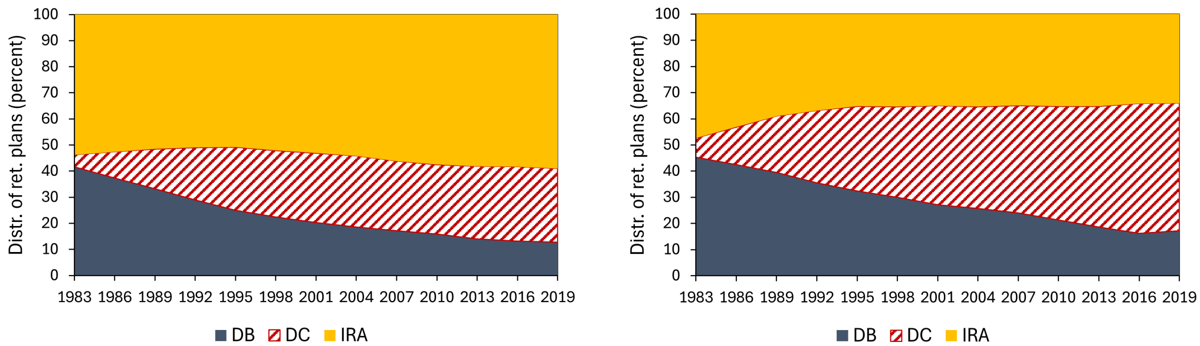
	1983 or earlier		1985 or later	
Retirement plan	College	Non-College	College	Non-College
Defined benefit	45.3	41.5	16	13
Defined contribution	7.3	4.7	50	28
IRA	47.4	53.7	34	59

The payments received by defined benefit retirees are based on the defined benefit function in Scholz, Seshadri, and Khitatrakun (2006). Their regression depends on union status, years of service in the pension-covered job, and expectations about earnings in the last year of work.

SCF data for years 1983 and 1989-2019 determines the distribution of the different types of retirement plans. The distribution depends on education and changes over time, see Figure 4. A respondent in the survey is deemed a defined benefit worker if he has at least one defined benefit retirement plan. The distribution in year 1986 is assumed to be an average of years 1983 and 1989. A rolling average using three years is applied to smooth the data.

There was a policy change for retirement plans in 1982 when defined contribution retirement plans were introduced into the economy. I assume the distribution of retirement plans for all cohorts entering the model prior to the policy change are equal to one another. Similarly, I assume the distribution of retirement plans for all cohorts entering the model after the policy change are equal to one another. The earliest survey data containing information on the distribution of retirement plans is 1983. I assume the distribution of retirement plans for all cohorts entering the model in 1983 or earlier is equal to the distribution in 1983. By 2019 almost all cohorts entering the economy before the policy change has exited the labor market. I therefore assume the distribution observed in the data in 2019 is equal to the distribution of retirement plans for cohorts entering the economy after the policy change. The distribution for cohorts entering the model after 1983 is therefore assumed equal to the distribution for year 2019. Table 2 reports the shares of each type of retirement plan when the cohorts enter the model.

Cohorts entering the model in 1983 or earlier are subject to a retirement plan shock.



(a) Non-college workers

(b) College workers

Figure 4: Distribution of types of retirement plans for college and non-college workers.

Defined benefit and IRA workers in these cohorts face a probability of becoming a defined contribution worker next period. Only workers younger than 55 years face the shock. The probabilities depend on the type of retirement plan and the time period. The probability distribution of the shock is uniform. The size of the shock is determined such that the probability distributions of retirement plans matches the distribution shown in Figure 4. The probability of changing plans is 0 for 1981 or earlier and for 2001 or later. The shocks are presented in Table 3.

Social Security benefits

This section will be updated shortly. The Social Security benefits SS_ζ are estimated by approximating the modelling in Huggett and Ventura (2000). Making their old-age component of Social Security benefits consistent with my model gives

$$\begin{aligned} SS_\zeta = {} & 0.9 \min(\zeta, 0.222) + 0.32 \max\left(0, \min(\zeta, 1.340) - 0.222\right) \\ & + 0.15 \max\left(0, \min(\zeta, 2.658) - 1.340\right). \end{aligned} \tag{4}$$

Labor productivity

The deterministic life cycle component and the stochastic component of earnings is estimated following Heathcote, Storesletten, and Violante (2010) closely.

The deterministic life cycle component w_{je} is based on the following ordinary least-square regression: The logarithm of earnings is regressed on a time dummy, a time dummy

Table 3: Probability for defined benefit and IRA workers to become a defined contribution worker next period.

	College		Non-College	
	DB	IRA	DB	IRA
1981 and earlier	0.00	0.00	0.00	0.00
1983	0.02	0.07	0.05	0.00
1985	0.02	0.07	0.05	0.00
1987	0.03	0.07	0.05	0.00
1989	0.03	0.07	0.08	0.00
1991	0.03	0.00	0.08	0.00
1993	0.03	0.00	0.08	0.00
1995	0.03	0.00	0.08	0.00
1997	0.03	0.00	0.08	0.00
1999	0.00	0.00	0.08	0.00
2001 and later	0.00	0.00	0.00	0.00

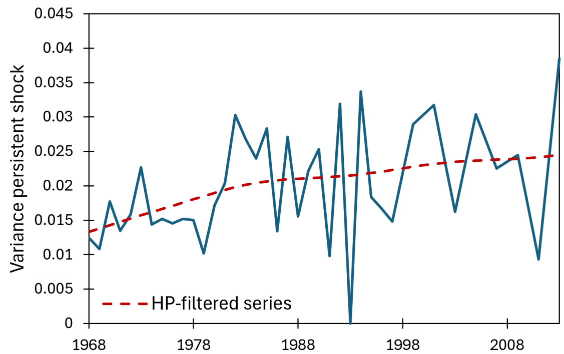


Figure 5: Estimates of the variance of the persistent earnings shock.

interacted with an education dummy, a cubic polynomial in age, and a cubic polynomial in age interacted with an education dummy

$$\ln v_{ijt} = \alpha_{0t} + \alpha_{1t} e_i + \alpha_2 j_{ijt} + \alpha_3 j_{ijt}^2 + \alpha_4 j_{ijt}^3 + \alpha_5 e_i j_{ijt} + \alpha_6 e_i j_{ijt}^2 + \alpha_7 e_i j_{ijt}^3 + u_{ijt}.$$

v_{ijt} is the earnings of individual i of age j at time t, e_i is the education dummy, and j_{ijt} is the age. α are the regression coefficients. The residuals u_{ijt} are estimates of the stochastic earnings component. It is estimated as the sum of two orthogonal components: a persistent autoregressive shock η_{ijt} and a transitory shock $\tilde{\varepsilon}_{ijt}$

$$u_{ijt} = \eta_{ijt} + \tilde{\varepsilon}_{ijt}$$
$$\eta_{ijt} = \rho \eta_{i(j-1)(t-1)} + \varepsilon_{ijt}.$$

ρ is the autocorrelation coefficient. $\tilde{\varepsilon}_{ijt}$ and ε_{ijt} are drawn from distributions with mean zero and variances $\sigma_{\tilde{\varepsilon}t}^2$ and $\sigma_{\varepsilon t}^2$ respectively. I assume η_{ijt}, $\tilde{\varepsilon}_{ijt}$, and ε_{ijt} are orthogonal to each other and i.i.d. across individuals in the population. The initial value of the persistent shock is drawn from a time-invariant distribution with mean zero and variance σ_η^2.

The parameters are estimated by minimizing the distance between the empirical and theoretical covariance moments of the stochastic earnings component. Individuals in the sample are grouped into 10 year adjacent age cells and the empirical moments are constructed using these cells. The identity matrix is used as the weighting matrix.

PSID data is used for the estimation. Data is available annually from 1967 to 1996 and biennially until 2018. For years prior to the first year the persistent shock is assumed to be equal to the value in the first year. For years with biennial missing data the persistent shock is assumed to be an average of the estimates in the year before and after. I estimate parameters up until 2012. Data for 2014, 2016, and 2018 are used to improve the estimates of parameters in earlier years only. For years with missing data, the autocorrelation coefficient and the variance of the initial persistent shock are assumed equal to their respective time-invariant values.

The autocorrelation coefficient is estimated to be 0.971. The variance of the initial persistent shock is estimated to be 0.179. Estimates of the persistent shock increases from 0.013 to 0.024 over the time period. A Hodrick-Prescott filter with smoothing parameter 800 is applied to the time series before the estimates are used in the model, see Figure 6. In the model, parameters in periods prior to the first and after the last period of estimates are assumed equal to the parameters in the first and last period respectively. The transitory shock is assumed to be fully insurable in the model.

Taxes

The main reference for the estimation of the tax functions is Heathcote, Storesletten, and Violante (2017). Let $T(y)$ be the tax at income level y. The tax function is given by

$$T(y) = y - (1 - \tau_0)y^{1-\tau_1}.$$

Let $\tilde{y}$ be the income after tax, e.g. $\tilde{y} = y - T(y)$. The relationship between income before and after tax is

$$\tilde{y} = (1 - \tau_0)y^{1-\tau_1}, \tag{5}$$

which is used to estimate the tax parameters. The tax functions for workers and retirees are estimated separately as there are differences in how these groups are taxed.[4] One important difference is Social Security benefits: While Social Security benefits are included in a retiree's income before tax y, the share of the payments that is taxable by law depends on the total income of the retiree.[5]

4. For more information on taxation of seniors and retirees, see IRS website https://www.irs.gov/individuals/seniors-retirees.

5. See information on tax treatment of Social Security payments on Social Security Administration website https://www-origin.ssa.gov/benefits/retirement/planner/taxes.html.

SCF data from 1988 to 2018 is used to estimate the tax functions. The survey includes financial information on the respondents' retirement plans. NBER's TAXSIM program is used to estimate the tax paid by each record in the dataset, see Feenberg and Coutts (1993). Moore's code is used to prepare the data for TAXSIM.[6]

Income before tax includes earnings, Social Security benefits, defined benefit payments, withdrawals from retirement accounts, business income, interest, dividend, capital gains, rent, royalties, child support and alimony, and other income. The taxable deductions mortgage interests and charitable contributions[7] are subtracted from the income measure. At last, employer's share of FICA tax is added to arrive at the final measure of income before tax.

Two remarks are in order to give further insight into how Moore prepares the data for TAXSIM. First, the tax rate of capital gains depends on how long an asset is held, but this information is not recorded in the SCF. Capital gains are therefore split into short-term and long-term portions. IRS Statistics of income Individual report (table 1.4) gives information on the share of capital gains that are short-term and long-term on the aggregate level. The shares are given for intervals of adjusted gross income (AGI): less than 50,000 USD, between 50,000 and 100,000 USD, and more than 100,000 USD. The amount of short-term and long-term capital gains for a record in the SCF survey is estimated based on the AGI for that record.

Second, some couples file taxes separately, and their income is split by the following strategy: earnings, Social Security benefits, defined benefit payments, withdrawal from retirement accounts, and business income is given at an individual level and split accordingly. Interest, dividend, capital gains, and other income is given at the household[8] level and is therefore split in two. Mop up variables of Social Security and pension income and business income is also given at the household level, but is split based on the shares of the corresponding individual income measures. Child support and alimony is split based on information

6. I thank Kevin Moore for making available the SAS code on NBER website https://taxsim.nber.org/to-taxsim/scf27-32/. This section covers the main strategy on how the data is prepared for TAXSIM. Please see the code for more details.

7. The SCF only records charitable contributions of at least 500 USD.

8. The term household in this paper refers to the term primary economic unit in the SCF.

regarding former marriages.

After tax income equals income before tax minus federal taxes as estimated by TAXSIM. Tax incentives for saving for retirement is a main consideration of this paper and transfers are therefore not included in the tax functions: In the model, workers who save a large share of their income in the retirement asset might have a small taxable income. Nevertheless, they should not receive transfers just because their taxable income is small.

Only men are included in the analysis in this paper and therefore some adjustments are necessary for taxpayers who file jointly with a partner. Let y_{HH} and $T(y_{HH})$ be the household's income and taxes respectively. Similarly, let y_M and $T(y_M)$ be the man's income and taxes. $T(y_M)$ is estimated such that the following ratios are equal

$$\frac{T(y_{HH})}{y_{HH}} = \frac{T(y_M)}{y_M}. \tag{6}$$

The households income is a record in the SCF dataset and the household's taxes are estimated by TAXSIM. The man's income is estimated from the SCF dataset similarly to how couples filing separately is estimated to split their income. Equation (6) then gives the man's taxes which is used in the estimation of the tax functions. The final dataset used to estimate the tax functions includes both men filing jointly and men filing separately or individually.

The tax parameters are estimated by ordinary least squares regressions using the logarithmic form of equation (5). The parameters for workers and retirees are estimated by separate regressions on groups of workers and retirees. Respondents at least 65 years old and receiving Social Security benefits are deemed retirees while other respondents are deemed workers. The tax parameters for workers τ_0^W and τ_1^W are estimated to be 0.066 and 0.106 respectively. Similarly, the tax parameters for retirees τ_0^R and τ_1^R are estimated to be 0.058 and 0.073. The FICA tax rate is set to 15.3 percent.

Survival probability

Survival probability in the model depends on age, education, and cohort while survival probability published by the Social Security Administration depends on age and cohort only. To allow survival probability to also depend on education, I follow Attanasio, Kitao, and Violante (2010) and Falcettoni and Nygaard (2023) and assume the college premium in survival probability $\Delta\Psi_j$ depends on age, but is constant across time. The college premium is the difference between survival probability for college and non-college individuals

$$\Delta\Psi_j = \mathbb{1}_{e=c}\psi_{jet} - \mathbb{1}_{e=nc}\psi_{jet}. \tag{7}$$

ψ_{jet} is the survival probability at age j, education level e, and time t. The indicator functions $\mathbb{1}_{e=c}$ and $\mathbb{1}_{e=nc}$ equals one for college and non-college individuals respectively. The combined survival probability Ψ_{jt} for a given age at a given time is the weighted average of the survival probability for a given age, education, and time group

$$\Psi_{jt} = (1 - \Lambda^c_{jt})\mathbb{1}_{e=nc}\psi_{jet} + \Lambda^c_{jt}\mathbb{1}_{e=c}\psi_{jet}. \tag{8}$$

Λ^c_{jt} is the share of college educated individuals of age j at time t. Combining equation 7 and 8 gives survival probability ψ_{jet} as an expression of the combined survival probability, education premium, and share of college educated individuals. The latter three can be obtained in the literature or estimated using data and then used to obtain an estimate of the survival probability depending on age, education, and time.

Bell and Miller (2005) publishes estimates of survival probability given age and cohort, e.g. the combined survival probability Ψ_{jt}. The estimates used in this analysis are the cohort life tables and the first cohort in the study is born in 1900. Survival probability for cohorts born prior to 1900 are assumed to be equal to the 1900 cohort.

The education premium in survival probability is estimated using data from the National Vital Statistics System (NVSS) and CPS. NVSS records all deaths in the U.S. and CPS gives an estimate of the number of individuals alive. The education premium is estimated by calculating the survival probability for college and non-college individuals for all ages before taking the difference between the two education groups. Assuming a log-linear relationship I run an ordinary least-squares regression of the education premium on age. The results yields an estimate of the education premium.

The share of college educated individuals is estimated using CPS data. The relationship between the share of college educated individuals and age is assumed to be linear. The ordinary least-square regression of the share of college educated individuals on age for each cohort gives the estimate. The share of college educated individuals increases from 4.1

Table 4: Parameters determined jointly in equilibrium

Parameter	Description	Target	Value
β	Discount factor	Taxable wealth to taxable income = 1.018	0.903
φ_1	Bequest parameter	Taxable bequest to taxable assets = 0.080	-260.688
φ_2	Bequest parameter	-	30

percent for cohorts entering the model in 1925 or earlier to 22.6 percent for cohorts entering the model in 1989.

4.2 Parameters determined jointly in equilibrium

Table 4 reports the parameters determined jointly in equilibrium. The discount factor β is calibrated to 0.955 such that the taxable wealth to taxable income ratio is 1.018. Taxable wealth and taxable income is estimated using data from the SCF for years 1988-2018. Taxable income includes earnings and return on taxable assets. Taxable assets are recorded for the household as a whole. Assuming the ratio is equal for men and women, both wealth and income measures are estimated for the household as a whole.

The bequest parameters φ_1 is calibrated to -260.688. RAND HRS data from 1998 to 2018 is used to estimate the moment to calibrate the parameter. Bequest is defined as taxable assets the last period the respondent is alive. φ_1 is calibrated to match the ratio of total bequest to total taxable assets for men age 65 and above for each year. Taking the average across all years gives a ratio of 0.080. Taxable assets are recorded at the household level in the survey. If the household consists of a couple, the man is assumed to own half of the assets. The bequest parameters φ_2 is set to 30.

5 Results

The model's fit to the data is first assessed by comparing inequality measures in levels and changes over the time period. Figure 7 compares the $P90/P10$ and $P75/P25$ ratios from 1989 to 2019 in the data and in the model. The model reproduces the increase in inequality

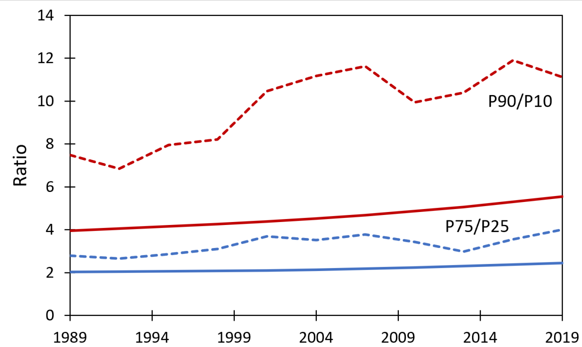


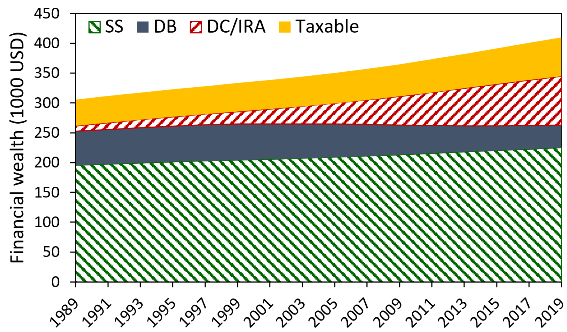
Figure 6: Ratios of percentiles of financial wealth for retirees from 1989 to 2019. The graphs show ratios in the data (dashed) and the model (solid).

over time, but understates the magnitude. The $P90/P10$ ratio rises by 1.58 in the model and by 3.64 in the data, so the model accounts for roughly 43 percent of the observed increase. The level of the ratio is similarly understated. The same pattern holds for the $P75/P25$ ratio, although the gap between the model and the data is smaller.

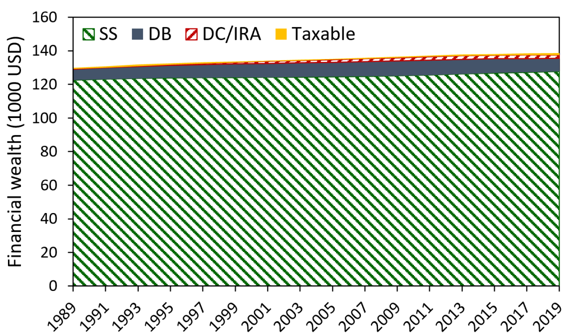
Figure 8 reports the model decomposition of financial wealth for retirees, the counterpart of the data in Figure 2. For the average measure, the model matches two features of the data: all four components of financial wealth are quantitatively important throughout the period, and defined contribution and IRA assets are the main source of the increase. The model nonetheless generates less total financial wealth than the data, overstating Social Security wealth and understating the remaining three components.

At the bottom of the distribution, Social Security accounts for almost all financial wealth in both the model and the data, and neither shows much change over the time period. The model overstates Social Security wealth for this group by roughly one third. One likely source of this discrepancy is the absence of unemployment in the model. Workers at the bottom of the distribution are more likely to experience unemployment, which reduces realized lifetime earnings and the resulting Social Security benefit.

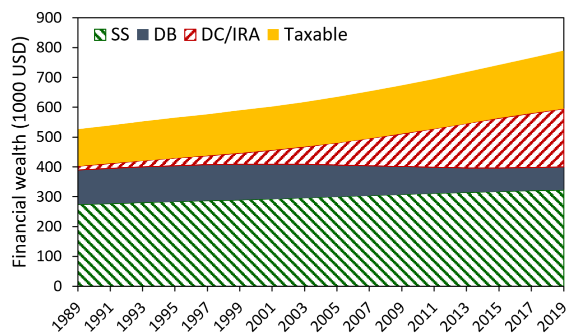
At the top of the distribution, the model matches the data on the composition of wealth and on the role of defined contribution and IRA assets as the main source of the increase. The level of total financial wealth is nonetheless lower in the model than in the data. Taken together with the bottom of the distribution, the model reproduces the qualitative pattern



(a) Average



(b) 5-15 percentiles



(c) 85-95 percentiles

Figure 7: Decomposition of financial wealth into Social Security (SS), defined benefit (DB), defined contribution and IRA (DC/IRA), and taxable (Taxable) assets. The graphs show different parts of the wealth distribution. Wealth is measured in 2019 dollars.

that the increase in inequality reflects a rise in financial wealth at the top with little change at the bottom. The overstatement of Social Security wealth at the bottom and the understatement of total wealth at the top together explain why the model captures the direction but not the magnitude of the increase in the $P90/P10$ ratio.

The next step measures how each channel contributes to the increase in inequality in financial wealth for retirees. In the full model, all three channels change over time as observed in the data. The contribution of a channel is measured by holding it constant at its 1973/74 level over the remainder of the time period. Figure 9 displays the $P90/P10$ ratio in the full model and as the channels are held constant one after another. The full model generates the largest increase in the ratio. As each additional channel is held constant, the ratio falls, reaching its lowest level in the baseline in which all three channels are held constant. The

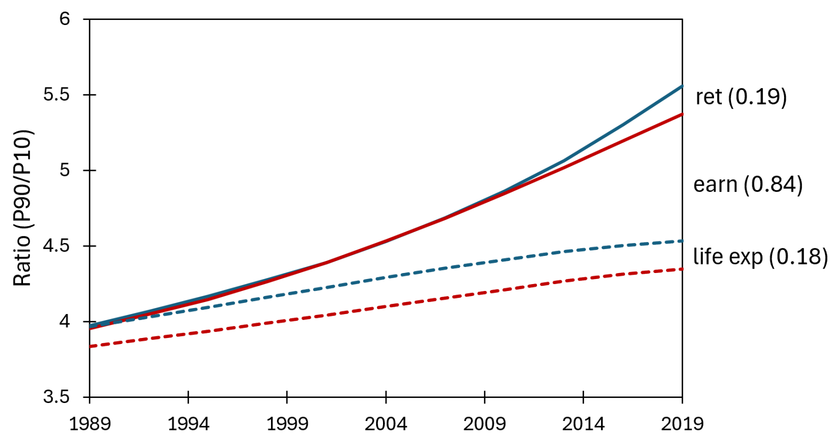


Figure 8: The results of allowing the channels to change or keeping them constant over the time period.

increase that remains in the baseline reflects aggregate technological change and changes that occurred before 1973/74 but affect inequality among retirees later in the time period.

First, holding the distribution of retirement plans constant at its 1973/74 level decreases the $P90/P10$ ratio by 0.19. IRAs were introduced into the U.S. economy in 1975, and before then a worker with neither a defined benefit nor a defined contribution plan had no tax-advantaged way to save for retirement. Holding this channel constant fixes the distribution of plan types at its 1973/74 level and does not introduce IRAs into the economy.

The three plan types differ in how a worker saves for retirement, and each shapes inequality differently. An IRA worker can save in the tax-deferred asset, which carries a more favorable tax treatment than the standard asset and so yields a higher discounted return. A defined contribution worker can also save in the tax-deferred asset and receives a proportional match on the amount saved, raising the return further. Both plans cap saving in the tax-deferred asset, with a higher limit for the defined contribution worker. Workers with low earnings do not find it optimal to save for retirement, so these incentives matter only for workers with high earnings. Because the tax-deferred asset yields a higher discounted return than the standard asset, a worker with high earnings saves more for retirement once they have access to it. And because the match raises that return further and the higher limit allows more saving, a defined contribution worker with high earnings accumulates more than a comparable IRA worker.

A defined benefit worker, by contrast, saves in a tax-advantaged way for retirement through mandatory contributions: a fixed share of earnings is subtracted every period and returned as a predetermined payment in retirement. This worker has no access to the tax-deferred asset. These contributions are mandatory for high and low earners alike. Low earners therefore save more for retirement than they would otherwise choose. High earners, however, save less than they would otherwise choose: their only tax-advantaged saving for retirement is the compulsory contribution to the defined benefit plan. If they wish to save more for retirement, any additional saving must be in the standard asset, which yields a lower return. A defined benefit worker with high earnings therefore accumulates less in total than if they had access to the tax-deferred asset.

Inequality in financial wealth is therefore lower among defined benefit workers than among defined contribution and IRA workers, both because low earners save more and because high earners save less. Shifting the population away from defined benefit plans and toward defined contribution and IRA plans therefore raises overall inequality in financial wealth for retirees. At the percentile level, allowing the distribution of plans to change raises the 90th percentile: as the share of defined contribution workers grows, high-earning workers who would otherwise have held a defined benefit plan accumulate more once they can save in the tax-deferred asset, raising wealth at the top of the distribution. The 10th percentile is unchanged. The bottom of the distribution consists of low earners who do not find it optimal to save for retirement and hold essentially only Social Security.

Second, additionally holding earnings inequality constant at the 1973/74 level decreases the $P90/P10$ ratio by another 0.84. Earnings inequality enters the model through three components, each of which increases over the time period: the variance of the idiosyncratic earnings shock, the college premium, and the share of workers with college education. A retiree's financial wealth comes from Social Security benefits, any defined benefit payment, and the assets they accumulated while working, each of which is shaped by earnings over the working life. Greater earnings inequality therefore raises inequality in financial wealth for retirees through three routes. Social Security benefits are a progressive function of average lifetime earnings, so a rise in earnings inequality widens the dispersion of benefits, though by less than one for one. Defined benefit payments, by contrast, are a linear function of earnings in a worker's final period of work and pass rising earnings inequality through directly. Optional saving for retirement transmits earnings inequality most strongly of the three: a worker at the bottom of the distribution does not find it optimal to save for retirement, whereas a worker with high earnings saves substantially more as their earnings rise. The response is strongest for a defined contribution worker, whose additional saving goes into the tax-deferred asset and earns the matching contribution, raising both the return and the amount accumulated. At the percentile level, allowing earnings inequality to vary raises the 90th percentile: workers there receive larger Social Security and defined benefit payments and save substantially more for retirement. The 10th percentile barely moves, since a worker there does not save for retirement and holds essentially only Social Security, whose benefit reflects average lifetime earnings that change little at the bottom of the distribution.

Third, additionally holding life expectancy constant at the 1973/74 level decreases the $P90/P10$ ratio by another 0.18, returning the model to the baseline in which all three channels are held constant. Holding this channel constant fixes the age- and education-specific survival probabilities at their 1973/74 level. When this channel is allowed to change over the time period, these probabilities rise for both college and non-college workers, and by more for college workers. The mechanism operates through the discounting of Social Security and defined benefit payments and through the optimal choice of saving for retirement. When the survival probability rises, the discounted value of Social Security payments increases, because a worker is more likely to be alive to receive them; the same holds for defined benefit payments. A worker who expects to be retired for a long time chooses to save more for retirement than one who does not. The overall increase in survival probabilities therefore raises inequality in financial wealth across cohorts, while the widening gap between college and non-college workers raises it within the same cohort. At the percentile level, allowing life expectancy to change increases financial wealth at the 90th percentile: workers receive a higher discounted value of their Social Security and defined benefit payments and save more for retirement. It also increases the 10th percentile, though only marginally: workers hold essentially only Social Security, and its discounted value rises as they live longer. Because the 90th percentile rises by more than the 10th, the $P90/P10$ ratio increases.

The diagonal of Table 5 reports the single effect of each channel: the change in the

Table 5: Effects on the $P90/P10$ ratio of allowing the channels to vary over the time period

	Ret	Earn	Life exp	All three
Ret	0.35	-0.06	-0.07	-0.03
Earn		0.99	-0.15	
Life exp.			0.18	

$P90/P10$ ratio when that channel alone is allowed to vary over the time period while the other two are held at their 1973/74 level. Allowing only the distribution of retirement plans to vary raises the ratio by 0.35, allowing only earnings inequality to vary raises it by 0.99, and allowing only life expectancy to vary raises it by 0.18. The off-diagonal entries measure how much a channel's effect changes once a second channel is also allowed to vary, equivalently how much the channels together depart from the sum of their separate effects. These entries are negative because the channels raise inequality through overlapping mechanisms, so two channels varying together raise the ratio by less than the sum of each varying alone. For example, letting retirement plans and earnings vary together raises the ratio by 0.06 less than the sum of their two single effects. Each row then sums to that channel's marginal contribution to the increase in the $P90/P10$ ratio reported in Figure 9: $0.35 - 0.06 - 0.07 - 0.03 = 0.19$ for retirement plans, $0.99 - 0.15 = 0.84$ for earnings, and 0.18 for life expectancy. Because the channels overlap, the contribution attributed to each depends on the order in which they are held constant, so the table separates the single effects, which do not depend on this order, from the off-diagonal entries that make the overlap visible.

6 Conclusion

This paper studies the increase in inequality in financial wealth for retirees in the U.S. between 1989 and 2019, using the $P90/P10$ ratio to quantify it. The ratio rose from 7.48 to 11.13 over the period, an increase of 3.64. The paper asks how three channels contributed to this increase: the shift from defined benefit toward defined contribution plans together with the introduction of IRAs, rising inequality in earnings, and rising and increasingly heterogeneous life expectancy. To measure their contributions, I use a discrete-time, partial equilibrium, overlapping generations life cycle model with heterogeneous agents, calibrated to the U.S. economy. The contribution of each channel is measured by comparing the change in inequality when the channel varies over time versus holding it constant at its 1973/74 level.

The change in the distribution of retirement plans, together with the introduction of IRAs, raises the $P90/P10$ ratio by 0.19. The plan types differ in how a worker saves for retirement. Under a defined benefit plan, saving for retirement is mandatory: a fixed share of earnings is set aside in a tax-advantaged way under the same rule for every worker. Under defined contribution and IRA plans, the worker instead chooses how much to save in the tax-advantaged asset. Among defined contribution workers, a low earner does not find it optimal to save for retirement, while a high earner saves more for retirement than they would under a defined benefit plan. Financial wealth is therefore more unequally distributed among defined contribution than among defined benefit workers. The shift toward these plans widens inequality in financial wealth for retirees.

Rising inequality in earnings raises the $P90/P10$ ratio by 0.84. A retiree's financial wealth reflects the earnings they received over their working life, so wider inequality in earnings produces wider inequality in financial wealth among retirees. Rising life expectancy, together with the widening gap between college and non-college workers, raises the ratio by 0.18. A worker who expects to be retired longer saves more for retirement, so longer and increasingly unequal life expectancy widens inequality in financial wealth among retirees.

The model reproduces an increase in the $P90/P10$ ratio of 1.58, compared with 3.64 in the data, accounting for roughly 43 percent of the observed increase. It reproduces the direction of the increase but not its full magnitude. Two features of the model explain the gap. Social Security wealth is overstated at the bottom of the distribution, while total financial wealth is understated at the top. Reducing these discrepancies is left to future work, for instance by introducing unemployment, which would lower realized lifetime earnings and the Social Security wealth of workers at the bottom of the distribution.

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